Qalibrated Systems Limited

Document No.	QSL-CAL-PROC-006
Revision / Date	1.0 — 26 June 2026
Approved By	Technical Manager

CALIBRATION PROCEDURE

Flow Meters (Liquid) — Gravimetric Method per EURAMET Calibration Guide No. 19 Principles

1. Scope & Application

This procedure applies to the calibration of liquid flow meters (including electromagnetic, turbine, and positive-displacement types) using the gravimetric (start-stop) method, applying the same mass-to-volume conversion principles set out in EURAMET Calibration Guide No. 19 for gravimetric volume determination.

2. Reference Standards & Traceability

STANDARD	DESCRIPTION	TRACEABILITY
Reference Weighing Tank / Balance	Collects and weighs the diverted flow over a measured time interval	Calibrated per QSL-CAL-PROC-001 (NAWI), ≲ 12 months
Reference Timer	Measures the collection interval, synchronised with the diverter mechanism	Calibrated, ≲ 24 months
Reference Thermometer	Process fluid temperature for density determination	Accredited lab, ≲ 24 months

3. Environmental Conditions

PARAMETER	REQUIREMENT
Flow Stability	Steady-state flow established and confirmed before each diversion/collection cycle
Fluid Temperature	Recorded at each test point for density correction; held within a stable band during collection
Pipework	Meter installed per manufacturer's required upstream/downstream straight-pipe lengths

4. Procedure

1. **Pre-Test Inspection.** Inspect the meter for damage, confirm correct installation orientation, and purge air from the system.
2. **Test Point Selection.** Select a minimum of 5 flow rates distributed across the meter's calibrated range, typically including minimum, transitional, and maximum flow rates.
3. **Establish Steady Flow.** At each test point, establish steady-state flow and allow the meter under test to stabilise before beginning collection.
4. **Diversion & Collection.** Divert the flow into the weighing tank for a measured time interval, recording the meter's indicated volume/totaliser reading at the start and end of the interval.
5. **Weigh Collected Mass.** Weigh the collected fluid and convert to volume using the fluid density at the measured temperature.
6. **Repeat Determinations.** Repeat each test point a minimum of 3 times to establish repeatability.
7. **Calculate Error.** Determine the error of the meter as the difference between the indicated volume and the reference (gravimetrically determined) volume, expressed as a percentage of the reference volume.

5. Uncertainty & Calculation

The combined standard uncertainty includes: the calibration uncertainty of the reference balance; the uncertainty of the timing interval; the repeatability of determinations at each flow rate; the uncertainty of the fluid density correction; and any leakage or evaporation losses during collection. These are combined per the GUM method to give the expanded uncertainty (k=2) of the error at each test point.

6. Acceptance Criteria

The meter passes calibration where the error at each test point, together with its uncertainty, falls within the manufacturer's accuracy class or the client's stated tolerance requirement across the calibrated flow range.

Prepared By	Reviewed By	Approved By
Technical Manager	Quality Manager	Managing Director